Background gradient

Why Study the History of Mathematics?

- Mathematics was invented by real people solving real problems
- Every formula you have learned has a **human story** behind it
- Ideas emerged **independently** across continents
- Understanding *where* math came from helps us understand *what* it is

Simplified world outline (rectangle placeholders for contine

Key Insight

Mathematics is not a fixed body of knowledge handed down from above — it was created, debated, lost, and rediscovered by people just

Our Journey Today

Main timeline axis

This timeline is our roadmap. We will mark "you are here" as we travel through each era.

Where It All Began

Pre-Classical Mathematics ∼20,000 BCE – 500 BCE

Region highlights

Diagram

The Ishango Bone — The Oldest Mathematics?

Ishango Bone

Date: c. 20,000 BCE
Discovered: 1960, near Lake Edward, modern DR Congo
Now held at: Royal Belgian Institute of Natural Sciences, Brussels

Photograph of the Ishango bone showing three columns of tally marks (Royal Belgian Institute of Natural Sciences)

- A bone tool with notched markings
- Possibly represents: arithmetic sequences, prime numbers, or a **lunar calendar**
- One of the **earliest known mathematical artefacts**

Complex diagram

Babylon — The First Mathematicians

- Civilization in Mesopotamia (modern Iraq), c. 2000–500 BCE
- Developed a base-60 (sexagesimal) number system
- Why **60 minutes** in an hour, **360 degrees** in a circle
- Wrote mathematics on **clay tablets** in cuneiform script
- Thousands of mathematical tablets survive

Photograph of a cuneiform clay tablet with mathematical content (Yale Babylonian Collection or British Museum)

Wedge for 1

Key Insight

The next time you check the clock, thank

Plimpton 322 — A 3,800-Year-Old Mystery

Anonymous Babylonian Scribe(s)

Date: c. 1800 BCE
Origin: Mesopotamia (modern Iraq)
Now held at: Columbia University Rare Book & Manuscript Library

High-resolution photo of Plimpton 322 tablet (Columbia University, public domain)

- Contains Pythagorean triples
- Written ∼**1,200 years BEFORE Pythagoras**
- Interpretation debated:
  - Neugebauer & Sachs (1945): table of triples

a	b	c
119	120	169
3367	3456	4825
65	72	97

Selected rows from Plimpton 322

Egyptian Mathematics — Practical Genius

- Driven by **practical needs**:
  - Land surveying after Nile floods
  - Architecture (pyramids!)
  - Taxation and accounting
- Two main sources:
  - Rhind Papyrus (next slide) — larger, more complete
  - Moscow Papyrus — contains the frustum formula
- Used unit fractions: $\frac{2}{5} = \frac{1}{3} + \frac{1}{15}$

Section of the Rhind Papyrus (British Museum, EA 10057-8)

height

Ahmes — The Oldest Named Mathematician

Ahmes (Ahmose)

Dates: fl. c. 1550 BCE
Origin: Ancient Egypt
Contribution: Scribe who copied (and possibly compiled) the *Rhind Mathematical Papyrus* — our best source on Egyptian mathematics

Portion of the Rhind Papyrus showing a geometry problem (British Museum digital collection)

- Contains problems in **arithmetic**, **algebra**, **geometry**
- Implies $\pi \approx 3.16$ (via circle area formula)
- Ahmes wrote: *"Accurate reckoning for inquiring into things, and the knowledge of*

Egyptian Geometry in Action

Circle area (Rhind Papyrus):

Square 9x9 conceptual (scaled to 4.5)

Egyptian rule: $A \approx \left(\frac{8}{9}d\right)^2$
$\Rightarrow \pi \approx \frac{256}{81} \approx 3.16$

Frustum volume (Moscow Papyrus):

Frustum (trapezoid cross-section)

$V = \frac{h}{3}(a^2 + ab + b^2)$

Exactly correct!

c. 1850 BCE — from the Moscow Papyrus

Key Insight

Mesopotamian Algebra

Babylonians solved quadratic equations

(in verbal/geometric form, not symbolic notation)

Babylonian word problem:

"I have added the area and the side of my square: 0;45."

Modern equivalent:

$x^2 + x = \frac{3}{4}$

Babylonian solution (completing the square):

$x = \sqrt{\frac{3}{4} + \frac{1}{4}} - \frac{1}{2} = \frac{1}{2}$

*Adding the side: rectangle x*1*

Key Insight

Your algebra homework has a 4,000-year pedigree.

Think-Pair-Share: "Why do you think the Babylonians never developed

What Was Missing?

Ancient mathematics was computational, not axiomatic.

Diagram

vs.

Diagram

This is **not a deficiency** — the procedural approach served its purpose brilliantly for millennia. But the Greeks would add something new: the idea that you must *prove* your claims.

Mayan Mathematics — An Independent Tradition

Maya Civilization

Era: c. 4th century CE and earlier
Region: Mesoamerica (modern Mexico, Guatemala, Belize, Honduras)

- Vigesimal (base-20) number system
- Independently invented **the concept of zero**
- Long Count calendar required sophisticated positional arithmetic
- Zero used **centuries before Europe** knew the concept

Mayan Numerals

Mayan zero glyph and section of a Long Count calendar stone (Wikimedia Commons)

Legacy of Ancient Mathematics

What survives from the ancient world:

- **Base-60 system** $\rightarrow$ clocks, angles
- **Basic algebra** $\rightarrow$ quadratic equations
- **Geometry** $\rightarrow$ surveying, architecture
- **Approximations of** π
- **Independent number systems** across continents
- **Zero** appearing in multiple civilizations

Clock icon

"But one civilization would transform math from a tool into a way of thinking..."

The Greek Revolution

Math Becomes Logic ~600 BCE – 300 CE

Mediterranean outline hint

Diagram

Thales of Miletus

Thales of Miletus (Thalēs ho Milēsios)

Dates: c. 624 – c. 546 BCE
Origin: Miletus, Ionia (modern Turkey)
Contribution: Traditionally regarded as the first person to use *deductive reasoning* in mathematics

- Thales' theorem: any angle inscribed in a semicircle is a right angle
- Credited with predicting a solar eclipse
- Marks the shift from "*what*" to "*why*"

Semicircle

Map showing Miletus on the coast of Ionia (modern Turkey)

Pythagoras and the Pythagoreans

Pythagoras of Samos (Pythagoras ho Samios)

Dates: c. 570 – c. 495 BCE
Origin: Samos, Greece; active in Croton, Magna Graecia (southern Italy)
Contribution: Founded a school that saw numbers as the foundation of reality

- $a^2 + b^2 = c^2$ bears his name (but known earlier — recall Plimpton 322)
- Discovery that $\sqrt{2}$ is irrational — a crisis for their philosophy

"All is number"

Right triangle

$16 + 9 = 25 \quad \checkmark$

Euclid — *The Elements*

Euclid of Alexandria (Eukleides)

Dates: fl. c. 300 BCE (birth/death unknown)
Origin: Active in Alexandria, Ptolemaic Egypt
Contribution: Wrote *The Elements* — the most influential mathematics textbook in history

- Used for **over 2,000 years** as a textbook
- Organized geometry into a deductive system from axioms
- Proved the **infinitude of prime numbers**
- 13 books covering geometry, number theory, proportion

Page from a medieval manuscript of The Elements (Bodleian Library or Vatican Library)

Diagram

Archimedes — The Greatest?

Archimedes of Syracuse

Dates: c. 287 – 212 BCE
Origin: Syracuse, Sicily (Greek colony)
Contribution: Calculated π with high accuracy; discovered principles of buoyancy and the lever; method of exhaustion

- Bounded π: $3\frac{10}{71} < \pi < 3\frac{1}{7}$
- Used inscribed / circumscribed polygons
- **Method of exhaustion** $\rightarrow$ compared to early integral calculus
- Killed during the siege of Syracuse (212 BCE)

Circumscribed hexagon

$$\underbrace{P_{\text{in}}}_{\text{lower}} < \underbrace{C}_{\pi} < \underbrace{P_{\text{out}}}_{\text{upper}}$$

Used 96-sided polygons!

The Archimedes

Eratosthenes — Measuring the Earth

Eratosthenes of Cyrene

Dates: c. 276 – c. 194 BCE
Origin: Cyrene (modern Libya); worked in Alexandria
Contribution: Calculated Earth's circumference; Sieve of Eratosthenes for primes

Earth circle

- Compared shadow angles at **Alexandria** and **Syene** (Aswan)
- At summer solstice: no shadow in Syene, $\sim 7.2^\circ$ shadow in Alexandria
- Result: $\sim$**252,000 stadia** $\approx$

Apollonius — Conic Sections

Apollonius of Perga

Dates: c. 262 – c. 190 BCE
Origin: Perga, Pamphylia (modern Turkey); worked in Alexandria
Contribution: Wrote *Conics* — named the ellipse, parabola, and hyperbola

- The definitive ancient work on conic sections
- These names are his: ellipse, parabola, hyperbola
- Essential for Kepler's planetary orbits

Draw cone outline

Slicing a cone at different angles produces all four curves.

Diophantus — Father of Algebra?

Diophantus of Alexandria

Dates: fl. c. 250 CE
Origin: Alexandria, Roman Egypt
Contribution: Wrote *Arithmetica* — algebraic problems seeking integer or rational solutions; introduced proto-algebraic notation

- Diophantine equations (integer solutions) named after him
- One of the earliest uses of **symbolic abbreviations** for unknowns
- Fermat's "Last Theorem" was *scribbled in*

Page from a manuscript of Arithmetica (Wikimedia Commons)

Diagram

Hypatia of Alexandria

Hypatia of Alexandria (Hypatia hē Alexandrinē)

Dates: c. 350–370 – 415 CE
Origin: Alexandria, Roman Egypt
Contribution: Mathematician, astronomer, philosopher. Commentaries on Diophantus and Apollonius. Head of the Neoplatonist school in Alexandria.

- Wrote commentaries on Diophantus's *Arithmetica* and Apollonius's *Conics*
- One of the **last great scholars** of ancient Alexandria

"Hypatia" by Charles William Mitchell, 1885 (Laing Art Gallery, Newcastle)

Diagram

Greek Legacy and the Great Transition

What the Greeks gave us:

- Axiomatic reasoning and **proof**
- The idea that math is about *eternal truths*, not just practical calculation
- Geometry as a deductive system
- Foundations of number theory

But then...

- Much Greek knowledge **lost** in the West
- **Preserved** in Byzantine and Islamic libraries
- Two very different journeys ahead...

Greek source

Greek texts survived thanks to Islamic and Byzantine scholars.

India

The Invention of Nothing (and Everything)

$\sim$500 BCE – 1200 CE

The number that changed everything

Early Indian Mathematics — The Sulba Sutras

- **Sulba Sutras** (c. 800–200 BCE)
- Geometric rules for Vedic altar construction
- Include **Pythagorean triples** — independently of Babylon and Greece
- Approximation of $\sqrt{2}$:

$$\sqrt{2} \approx 1 + \frac{1}{3} + \frac{1}{3 \cdot 4} - \frac{1}{3 \cdot 4 \cdot 34} \approx 1.4142156\ldots$$

(true value: $1.4142135\ldots$)

Falcon-shaped altar (simplified)

Diagram from the Sulba Sutras showing altar construction with geometric proportions

Key Insight

Pythagorean triples appear in at least THREE independent traditions: Babylo

Aryabhata

Aryabhata (Aryabhata I)

Dates: 476 – 550 CE
Origin: Kusumapura (likely modern Patna), Gupta Empire, India
Contribution: Wrote *Aryabhatiya* (499 CE) — trigonometric tables, place-value system, approximation of π, Earth's rotation

Statue of Aryabhata (IUCAA Pune) or page from a manuscript of Aryabhatiya (Wikimedia Commons)

- $\pi \approx 3.1416$ — remarkably accurate
- Used a place-value number system
- Tabulated **sine values** (the first systematic trigonometric table)

Data points

Brahmagupta — The Rules for Zero

Brahmagupta

Dates: 598 – c. 668 CE
Origin: Bhinmal, Rajasthan, India
Contribution: *Brahmasphutasiddhanta* (628 CE) — first to formalize rules for zero and negative numbers; general quadratic formula; cyclic quadrilateral formula

- First rules for arithmetic with zero:
  $a + 0 = a, \quad a \times 0 = 0, \quad 0 + 0 = 0$
- Negative $\times$ negative $=$ **positive**
- General solution to the **quadratic equation**

Rules in boxes

Cyclic quadrilateral

Bhaskara II (Bhaskaracharya)

Bhaskara II — "Bhaskara the Teacher"

Dates: 1114 – 1185 CE
Origin: Bijjada Bida (likely modern Karnataka), India
Contribution: Wrote *Lilavati* (mathematics) and *Bijaganita* (algebra); solved Pell's equation before Pell; preliminary concepts related to infinitesimals

Opening verse of Lilavati in Devanagari with English translation

- *Lilavati* — a math textbook addressed to his daughter
- One of the earliest textbooks written for a specific student
- Solved $x^2 - ny^2 = 1$ ("Pell's equation")

Diagram

The Kerala School — Calculus Before Calculus?

Madhava of Sangamagrama

Dates: c. 1340 – c. 1425 CE
Origin: Sangamagrama (modern Irinjalakuda), Kerala, India
Contribution: Founder of the Kerala school; infinite series for π, sine, and cosine — BEFORE European discovery

Also: **Nilakantha Somayaji** (1444–1544 CE, Kerala) — wrote *Tantrasamgraha* (1501), extended Madhava's work.

- Infinite series for π (Madhava–Leibniz):

Kerala school

Key Insight

Was this transmitted to Europe or independently rediscovered? An open question in the history of mathematics.

Indian Mathematics — A Legacy for the World

India gave the world:

- The decimal place-value system
- **Zero** as a number
- **Negative numbers** with formal rules
- Trigonometric functions (**sine**)
- **Infinite series**
- Solutions to Pell's equation
- The foundations of the number system we use today

Indian numerals

The numeral system you use every day was invented in India, transmitted through the Islamic world, and adopted by Europe.

"Meanwhile, on the other side of Asia, a parallel mathematical tradition was flourishing..."

China

Parallel Discoveries

$\sim$200 BCE – 1300 CE

Horizontal rods for 1-5

The Nine Chapters on the Mathematical Art

Jiuzhang Suanshu

Date: Compiled c. 2nd century BCE – 1st century CE
Origin: Han Dynasty, China
Content: 246 problems covering arithmetic, algebra, geometry, engineering

- China's most important ancient mathematical text
- Includes Gaussian elimination — ∼2,000 years before Gauss!
- Methods for **square roots** and **cube**

Page from the Nine Chapters showing mathematical problems (Wikimedia Commons / Chinese library digital collections)

Ground line

Liu Hui

Liu Hui

Dates: fl. c. 263 CE
Origin: Cao Wei kingdom, China (Three Kingdoms period)
Contribution: Commentary on the *Nine Chapters* (263 CE); polygon method for π; principles of cartographic surveying

- Used a polygon method (like Archimedes) to calculate π
- Achieved $\pi \approx 3.14159$ (5 decimal places)
- Developed independently of

Circle

Who	When	π accuracy
Archimedes	c. 250 BCE	$3.1408\ldots$
Liu Hui	263 CE	3.14159
Zu Chongzhi	c. 480 CE	$3.1415926\ldots$

Zu Chongzhi — The World's Best π for a Millennium

Zu Chongzhi

Dates: 429 – 500 CE
Origin: Liu Song / Southern Qi dynasties, China
Contribution: Calculated π to 7 decimal places — the most accurate value in the world for nearly 1,000 years

- $3.1415926 < \pi < 3.1415927$
- The fraction $\frac{355}{113}$ is accurate to **6 decimal places**
- This approximation was not bettered for

Diagram

Entries

Chinese Mathematics — Legacy

Independent discoveries:

- Gaussian elimination (∼2,000 yrs before Gauss)
- **Horner's method** for polynomials
- **Yang Hui's triangle** (= Pascal's triangle)
- **Chinese Remainder Theorem** — still used in modern cryptography

Triangle rows

Yang Hui published this 392 years before Blaise Pascal.

Key figures:

Sunzi (fl. c. 3rd–5th c. CE) — Chinese Remainder Theorem

Yang Hui (c. 1238–1298) — described the triangle in 1261

Session 1 Complete

"Greek knowledge is about to take two very different journeys. . ."

The House of Wisdom

When Baghdad Was the Center of the World

800–1400 CE

Diagram

The Translation Movement

- Under the Abbasid Caliphate, a massive translation project preserved Greek, Indian, and Persian texts in Arabic
- The **House of Wisdom** (*Bayt al-Hikma*) in Baghdad was the epicenter
- Without this effort, much of Greek mathematics might have been permanently lost

Greek Texts (some lost originals)

Key Insight

When Europeans later "rediscovered" Greek mathematics, they were often translating *Arabic* translations of Greek originals that no

Muhammad ibn Musa al-Khwarizmi — The Father of Algebra

Al-Khwarizmi

Full name: Abu Abdallah Muhammad ibn Musa al-Khwarizmi
(Arabic name — see manuscript image)

Dates: c. 780 – c. 850 CE
Origin: Khwarezm (modern Khiva, Uzbekistan);
worked in Baghdad at the House of Wisdom

- Wrote *Kitab al-Jabr wa-l-Muqabala* (c. 820)
- The word **"algebra"** comes from *al-jabr* ("completion")

Arabic script: al-jabr – requires Arabic font image

Page from the al-Jabr manuscript (Bodleian Library MS. Huntington

Al-Khwarizmi's Algebra: Completing the Square

His words (paraphrased):

"A square and ten roots equal thirty-nine dirhams"

Modern form: $x^2 + 10x = 39$

Geometric method:

Completing the square – geometric diagram

Step-by-step:

1. Start with $x^2 + 10x = 39$
2. Draw the x^2 square
3. Distribute $10x$ as two rectangles of $\frac{5}{2} \times x$
4. **Complete** the square: add $\left(\frac{5}{2}\right)^2 = \frac{25}{4}$
5. Now: $\left(x + \frac{5}{2}\right)^2 = 39 + \frac{25}{4} = \frac{181}{4}$
6. $x + \frac{5}{2} = \frac{\sqrt{181}}{2}$, so $x = \frac{\sqrt{181}-5}{2} = 3$

Key Insight

He solved the same equations you solve — but geometrically, in words, with no symbols.

Thabit ibn Qurra — Translator and Innovator

Thabit ibn Qurra

(Thabit ibn Qurra)

Dates: 826 – 901 CE
Origin: Harran (modern Turkey); worked in Baghdad

- Prolific translator: Euclid, Archimedes, Apollonius
- Extended Euclid's number theory
- Proved a generalization of the Pythagorean theorem to arbitrary triangles
- Discovered a rule for finding **amicable**

Arbitrary triangle

Manuscript page from Thabit's translation of Euclid

Abu Kamil — Extending the Algebra

Abu Kamil Shuja ibn Aslam

(Abu Kamil Shuja ibn Aslam)

Dates: c. 850 – c. 930 CE

Origin: Egypt

- Extended al-Khwarizmi's algebra *significantly*
- Worked with **irrational numbers** as both solutions *and* coefficients
  - Al-Khwarizmi had restricted to rationals
- Solved systems of equations
- Influenced al-Karaji and, through Latin

A problem from Abu Kamil's Algebra with its geometric solution

Al-Khwarizmi

Al-Karaji — Algebra Freed from Geometry

Abu Bakr al-Karaji

(sometimes written al-Karkhi)

Dates: c. 953 – c. 1029 CE

Origin: Karaj (or Karkh, Baghdad); worked in Baghdad

Visual representation: stacked squares

- Freed algebra from geometry — treated algebraic operations **purely arithmetically**
- Proved the **binomial theorem**
- Proved the sum-of-cubes formula using a method akin to **mathematical induction**
- Extended algebra to powers beyond cubes

Key Insight

He arguably used the first form of **mathematical induction** — a proof technique you may encounter at university.

Omar Khayyam — Poet and Mathematician

Omar Khayyam

Ghiyath al-Din Abu l-Fath Umar ibn Ibrahim al-Khayyam

Dates: 1048 – 1131 CE
Origin: Nishapur, Khorasan (modern Iran)

- Classified and solved **all cubic equations** geometrically (intersecting conic sections)
- Contributed to the **Jalali calendar** reform

Parabola diagram

Ibn al-Haytham — The First Scientist?

Ibn al-Haytham (Alhazen)

Abu Ali al-Hasan ibn al-Hasan ibn al-Haytham known in Latin as Alhazen

Dates: c. 965 – c. 1040 CE
Origin: Basra (modern Iraq); worked in Cairo

- *Kitab al-Manazir* (Book of Optics) — revolutionized the understanding of vision and light
- Solved **Alhazen's problem**: finding the reflection point on a spherical mirror (4th-degree equation)

Spherical mirror

Diagram from the Book of Optics (many historical

Al-Biruni — The Polymath [EXTENDED]

Abu Rayhan al-Biruni

(Abu Rayhan al-Biruni)

Dates: 973 – c. 1048 CE
Origin: Kath, Khwarezm (modern Uzbekistan)

- Calculated the Earth's radius to within $\sim$**1%** of the modern value
- Used trigonometry and a mountain
- Wrote *Tarikh al-Hind* on Indian mathematics and astronomy
- Pioneered comparative anthropology

Earth (partial arc)

Al-Samawal & Sharaf al-Din al-Tusi [EXTENDED]

Al-Samawal

(al-Samawal ibn Yahya al-Maghribi)

Dates: c. 1130 – c. 1180 CE
Origin: Baghdad

- Advanced algebraic notation
- First statement of **polynomial long division**

Diagram

Sharaf al-Din al-Tusi

(Sharaf al-Din al-Tusi)

Dates: c. 1135 – 1213 CE
Origin: Tus (modern Iran)

- Analyzed cubic equations by finding their **maxima and minima**
- Anticipating differential calculus!
- *Not* the same person as Nasir al-Din al-Tusi

Nasir al-Din al-Tusi — Trigonometry Becomes Its Own Field

Nasir al-Din al-Tusi

Muhammad ibn Muhammad ibn al-Hasan al-Tusi
known as Nasir al-Din al-Tusi

Dates: 1201 – 1274 CE
Origin: Tus, Khorasan (modern Iran)

- Wrote *Treatise on the Quadrilateral* — the **first work** to treat trigonometry as an independent discipline
- Stated and proved the **law of sines** for plane and spherical triangles

Triangle

Reconstruction of the Maragheh observatory

Jamshid al-Kashi — Master of Computation

Jamshid al-Kashi

Ghiyath al-Din Jamshid Mas'ud al-Kashi

Dates: c. 1380 – 1429 CE
Origin: Kashan (modern Iran);
worked at the Samarkand observatory under Ulugh Beg

- Calculated π to **16 decimal places** — a world record that stood for $\sim$180 years
- Invented an iterative algorithm for computing nth roots

Diagram

The Ulugh Beg observatory in Samarkand (modern

Islamic Mathematics: Legacy and Transmission

Islamic mathematicians did not merely "preserve" Greek knowledge — they **transformed** and **extended** it:

- ✔ **Invented** algebra as a discipline
- ✔ **Advanced** trigonometry to independence
- ✔ **Developed** computational methods
- ✔ **Pushed** toward symbolic thinking
- ✔ **Extended** number theory, optics, geodesy

Simplified map outline (just positioning)

Key Insight

The Islamic mathematical tradition was es-

Session 2 Complete

"These ideas are about to reach Europe and ignite a revolution. . ."

Europe Awakens

Arabic Mathematics Crosses the Mediterranean

1200–1500 CE

Diagram

Fibonacci — Bringing Arabic Mathematics to Europe

Leonardo of Pisa (Fibonacci)

Full name: Leonardo Bonacci ("Fibonacci")

Dates: c. 1170 – c. 1240–50 CE

Origin: Pisa, Italy; traveled extensively in North Africa

- Wrote *Liber Abaci* (1202) — introduced the **Hindu-Arabic numeral system** to Europe
- The "Fibonacci sequence" appeared as a **rabbit population problem**
- Convinced European merchants to abandon

Sequence display

Page from Liber Abaci showing Hindu-Arabic numerals

Nicole Oresme — Graphs Before Descartes

Nicole Oresme

Dates: c. 1320 – 1382 CE
Origin: Normandy, France; worked in Paris

- Pioneered **graphical representations** of changing quantities ("latitude of forms")
- A precursor to **Cartesian coordinates** — 250 years before Descartes
- Proved the **harmonic series diverges**:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \infty$$

Oresme's version

Diagram

Blaise Pascal — Probability and the Triangle

Blaise Pascal

Dates: 1623 – 1662 CE
Origin: Clermont-Ferrand, France

- Co-founded **probability theory** (correspondence with Fermat, 1654)
- "Pascal's triangle" — known centuries earlier in China, India, and the Islamic world
- Built one of the first **mechanical calculators** (the Pascaline, 1642)
- Major philosopher and writer (*Pensées*)

Triangle rows

The Pascaline calculator (Musée des Arts et Métiers, Paris)

The Stage is Set

By 1500, the ingredients for a mathematical revolution were in place:

1. Indian and Islamic mathematics absorbed via Arabic translations
2. The **printing press** spreading knowledge (Gutenberg, c. 1440)
3. Universities growing across Europe
4. A new confidence in human reason (the Renaissance spirit)

"And then came the Renaissance, when mathematics exploded. . . "

Vertical timeline

The Modern Age Begins

Revolution in Symbol, Method, and Thought

1500–1700 CE

Cardano and the Cubic Equation — A Mathematical Drama

Gerolamo Cardano

(Latin: Hieronymus Cardanus)

Dates: 1501 – 1576 CE
Origin: Pavia, Italy

- Published the general solution to **cubic equations** in *Ars Magna* (1545)
- But who really discovered it?
- The solution forced the invention of **complex numbers** ($\sqrt{-1}$)

Arrows with labels

François Viète — Symbolic Algebra

François Viète

(Latin: Franciscus Vieta)

Dates: 1540 – 1603 CE
Origin: Fontenay-le-Comte, France

- Introduced systematic use of **letters** for *both* unknowns and known quantities
- Vowels for unknowns, consonants for knowns
- A major step toward **modern algebraic notation**

Medieval style

John Napier — Logarithms Change Everything

John Napier

Dates: 1550 – 1617 CE
Origin: Edinburgh, Scotland

- Invented **logarithms** (published 1614)
- Transformed computation: turned **multiplication into addition**
- Laplace: logarithms "doubled the life of astronomers" by halving their computational labor

Page from Napier's logarithm tables (1614) Mirifici Logarithmorum Canonis Descriptio

"Napier's bones" — his mechanical multiplication device

René Descartes — Geometry Meets Algebra

René Descartes

(Latin: Renatus Cartesius)

Dates: 1596 – 1650 CE
Origin: La Haye en Touraine, France; lived in the Dutch Republic

- Created **analytic geometry** — the merger of algebra and geometry
- Published in *La Géométrie* (1637)
- **Cartesian coordinates**: describe curves as equations, solve geometric problems algebraically

Grid lines

Pierre de Fermat — The Prince of Amateurs

Pierre de Fermat

Dates: 1601 – 1665 CE
Origin: Beaumont-de-Lomagne, France

- Co-founded **probability theory** (with Pascal, 1654)
- Developed early **differential calculus** methods (tangents, maxima/minima)
- Stated **Fermat's Last Theorem** in a marginal note (c. 1637)

Timeline

The Fermat–Pascal correspondence on probability (1654)

Isaac Newton — The Method of Fluxions

Sir Isaac Newton

Dates: 1642 – 1726/27 CE
Origin: Woolsthorpe-by-Colsterworth, England

- Co-invented **calculus** ("the method of fluxions"), developed independently during 1665–1666
- Published *Principia Mathematica* (1687) — used calculus to describe gravity and planetary motion
- Also: optics, reflecting telescope

Page from the Principia (Cambridge University Library)

Newton's notation:

$\dot{x}$ (fluxion of x)

$\ddot{x}$ (second fluxion)

Still used in physics today for time

Gottfried Wilhelm Leibniz — The Notation that Won

G. W. Leibniz

Dates: 1646 – 1716 CE
Origin: Leipzig, Germany;
worked in Hanover and Paris

- Co-invented calculus **independently** of Newton (published 1684–86)
- His **notation** is what we use today
- Invented a mechanical calculator (Stepped Reckoner)
- Pioneered **binary arithmetic**

Winner arrow

Key Insight

The Bernoulli Family — A Mathematical Dynasty

The Bernoulli family of Basel, Switzerland: at least **eight** notable mathematicians across three generations.

Jacob Bernoulli (1655–1705)

Basel, Switzerland

Bernoulli numbers • *Ars Conjectandi* (1713)

The **law of large numbers**

Johann Bernoulli (1667–1748)

Basel, Switzerland

Taught Euler • Brachistochrone problem

Generation 1

start

Calculus Invented — The Floodgates Open

With calculus, analytic geometry, and modern notation in place, mathematics had the tools for **explosive growth**.

tiny 1637

What was now possible:

- ✔ Describe motion and change precisely
- ✔ Compute areas, volumes, arc lengths
- ✔ Model planetary orbits
- ✔ Predict physical phenomena mathematically

What came next:

- → Euler: the most prolific mathematician ever
- → Lagrange & Laplace: mechanics and celestial theory
- → Gauss: the "Prince of Mathematicians"

The Age of Giants

When Mathematics Conquered the Physical World

1700–1800 CE

Diagram

Leonhard Euler — The Most Prolific Mathematician in History

Leonhard Euler

Dates: 1707 – 1783 CE
Origin: Basel, Switzerland;
worked in St. Petersburg and Berlin

- Published ∼866 papers and books — more than any other individual mathematician
- Standardized notation: e, i, π, $f(x)$, Σ
- Founded **graph theory** (Königsberg bridges)
- Foundational contributions to number

Five constants labeled

Four land masses

Joseph-Louis Lagrange [EXTENDED]

Joseph-Louis Lagrange

(born Giuseppe Lodovico Lagrangia)

Dates: 1736 – 1813 CE
Origin: Turin, Sardinia (modern Italy); worked in Berlin and Paris

- *Mécanique Analytique* (1788) — reformulated Newtonian mechanics using **pure analysis**
- Boasted: "no diagrams in this book"
- **Lagrange multipliers** — optimization with constraints

Diagram

Portrait of Lagrange

Pierre-Simon Laplace [EXTENDED]

Pierre-Simon Laplace

(marquis de Laplace)

Dates: 1749 – 1827 CE

Origin: Beaumont-en-Auge, Normandy, France

- *Mécanique Céleste* — applied calculus to celestial mechanics, completing Newton's program
- **Laplace transform** — essential tool in engineering and physics
- Major contributions to **probability**

Diagram

Portrait of Laplace

Carl Friedrich Gauss — *Princeps Mathematicorum*

Carl Friedrich Gauss

Dates: 1777 – 1855 CE
Origin: Braunschweig (Brunswick), Germany;
worked in Göttingen

- *Disquisitiones Arithmeticae* (1801) — foundational work in number theory
- Proved the **fundamental theorem of algebra**
- Discovered **non-Euclidean geometry** (unpublished!)

Bell curve (filled)

Regular 17-gon

The Century of Rigor and Revolution

1800–1900 CE

Galois · Riemann · Cantor · Noether · Ramanujan · Kovalevskaya

Evariste Galois — Genius Cut Short

Evariste Galois (1811–1832)

Origin: Bourg-la-Reine, France

Key work: Founded group theory. Proved there is no general algebraic formula for polynomials of degree 5 or higher, building on Niels Henrik Abel (1802–1829, Norway) who first proved the impossibility of solving the general quintic by radicals (Abel–Ruffini theorem, 1824). Galois went further: he characterised *exactly which* equations can be solved by radicals, using the group structure of their roots.

Symmetry group of a triangle

Bernhard Riemann — Curved Spaces and a Million-Dollar Question

Georg Friedrich Bernhard Riemann (1826–1866)

Origin: Breselenz, Kingdom of Hanover, Germany; worked in Göttingen

Key work:

- Riemannian geometry — curved spaces of arbitrary dimension, essential for Einstein's general relativity
- The Riemann integral
- The **Riemann hypothesis** (1859)

Flat grid

Georg Cantor — Infinity Has Sizes

Georg Ferdinand Ludwig Philipp Cantor (1845–1918)

Origin: St. Petersburg, Russia (German–Danish parentage); worked in Halle, Germany

Key work: Founded set theory. Proved that there are different sizes of infinity — the real numbers are "more infinite" than the natural numbers. This was so controversial that leading mathematicians tried to block his career.

Suppose we list all reals between 0 and 1

Emmy Noether — The Mother of Modern Algebra

Amalie Emmy Noether (1882–1935)

Origin: Erlangen, Germany; fled to the US in 1933

Key work:

- Noether's theorem (1918): symmetries in physics $\longleftrightarrow$ conservation laws
- Transformed abstract algebra — ring theory, ideal theory
- Denied a proper faculty position for years because she was a woman

Noether's Theorem

Portrait of Emmy Noether (Wikimedia Commons)

Srinivasa Ramanujan — The Self-Taught Genius

Srinivasa Ramanujan Aiyangar (1887–1920)

Origin: Erode, Tamil Nadu, India; worked at Cambridge

Key work: Self-taught mathematical genius. Produced thousands of results in number theory, infinite series, and continued fractions, many without formal proof. His notebooks continue to yield new mathematics a century later. Collaborated with G. H. Hardy at Cambridge.

Page from Ramanujan's notebooks (Cambridge University Library)

Diagram

Sofia Kovalevskaya — First Woman Doctorate in Mathematics

Sofia Vasilyevna Kovalevskaya (1850–1891)

Origin: Moscow, Russia; worked in Stockholm, Sweden

Key work:

- First woman to obtain a doctorate in mathematics in modern Europe (Göttingen, 1874)
- Major contributions to analysis, partial differential equations, and mechanics
- The Kovalevskaya top — a new

Portrait of Sofia Kovalevskaya (Wikimedia Commons)

Three known integrable cases

The 19th Century — An Overview

Five pillars

Transition: *"The 20th century would bring a crisis at the very foundations of mathematics. . ."*

Crisis, War, and Computers

1900–1950 CE

Hilbert · Gödel · Turing · von Neumann · Nash

David Hilbert — The Foundations Crisis

David Hilbert (1862–1943)

Origin: Königsberg, Prussia (modern Kaliningrad, Russia); worked in Göttingen

Key work: In 1900, posed 23 unsolved problems that shaped 20th-century mathematics. Hilbert's program aimed to put *all* of mathematics on a provably consistent, complete foundation.

G "odel's theorem in plain English

Kurt Gödel (1906–1978)

Origin: Brno, Austria-Hungary (now

Alan Turing — The Birth of Computer Science

Alan Mathison Turing (1912–1954)

Origin: London, England; worked at Cambridge, Bletchley Park, Manchester

Key work:

- The Turing machine (1936) — theoretical foundation of all computers
- Proved the **halting problem** is unsolvable
- Broke wartime Enigma at Bletchley Park, building on

Tape

John von Neumann — The Last Great Polymath

John von Neumann (1903–1957)

Born: Neumann János Lajos
Origin: Budapest, Hungary; worked in Germany and the USA (Princeton)

Key work: Contributions to quantum mechanics, game theory, computer architecture, cellular automata, nuclear weapons design, and set theory axiomatisation. Possibly the last great polymath.

CPU block

Key Insight

John Nash — Game Theory and the Nash Equilibrium

John Forbes Nash Jr. (1928–2015)

Origin: Bluefield, West Virginia, USA; worked at MIT and Princeton

Key work:

- Nash equilibrium (1950) — no player can improve their outcome by unilaterally changing strategy
- Applications: economics, evolutionary biology, AI
- Nobel Prize in Economics (1994)
- Struggled with schizophrenia for decades; eventually recovered

Matrix frame

Mathematics Goes Digital

1950–2000 CE

Shannon · Lorenz · Mandelbrot · Wiles

Cryptography · Computer-Assisted Proof

Claude Shannon — The Bit and Information Theory

Claude Elwood Shannon (1916–2001)

Origin: Petoskey, Michigan, USA; worked at Bell Labs and MIT

Key work: Founded information theory ("A Mathematical Theory of Communication," 1948). Defined the **bit** as the fundamental unit of information. Showed that all communication can be reduced to binary digits.

Blocks

Key Insight

Edward Lorenz — The Butterfly Effect

Edward Norton Lorenz (1917–2008)

Origin: West Hartford, Connecticut, USA; worked at MIT

Key work: Discovered deterministic chaos (1963) while running a weather simulation. Tiny rounding differences in initial conditions produced wildly different outcomes — the **butterfly effect**. Even perfectly deterministic systems can be practically unpredictable.

Approximate the butterfly shape with parametric curves

The Four-Color Theorem (1976) — Computers Enter Mathematics

Kenneth Appel (1932–2013) & Wolfgang Haken (1928–2022)

Proved at: University of Illinois, 1976

Any map can be coloured with at most four colours such that no two adjacent regions share a colour. Appel and Haken used a computer to check reducible configurations — the **first major theorem proved with essential computer assistance**.

A simple map requiring 4 colors

Benoit Mandelbrot — Fractals and Hidden Patterns

Benoit B. Mandelbrot (1924–2010)

Origin: Warsaw, Poland (Lithuanian-Jewish origin); worked in France and at IBM/Yale in the USA

Key work: Founded fractal geometry. The Mandelbrot set (1980) revealed infinite complexity from simple equations. Natural phenomena — coastlines, mountains, blood vessels, financial markets — have fractal structure

We use a simplified visual representation since true Mandelb

Cryptography — The Mathematics of Secrecy

Public-key cryptography, invented in the 1970s, made secure internet communication possible. Based on number theory — specifically the difficulty of **factoring large numbers**.

Diffie–Hellman Key Exchange (1976)

Whitfield Diffie (b. 1944, USA)
Martin Hellman (b. 1945, USA)

Co-invented public-key cryptography.

RSA Algorithm (1977)

Alice and Bob

Quick Poll

Raise your hand if you have used encryption today.

Quick Poll

Raise your hand if you have used encryption today.

Answer: everyone.

Diagram

Andrew Wiles — Fermat's Last Theorem Proved

Sir Andrew John Wiles (b. 1953)

Origin: Cambridge, England; worked at Princeton

Key work: Proved Fermat's Last Theorem in 1995 (after finding and fixing a gap in his 1993 proof). Used deep connections between elliptic curves and modular forms (the Taniyama–Shimura–Weil conjecture, now the modularity theorem). Closed a **358-year-old problem**.

The simple statement

Key Insight

The 20th Century — An Overview

Bottom summary

Mathematics in the Age of Machines

2000–2026 CE

Perelman · Tao · Mirzakhani · Uhlenbeck

Machine Learning · Large Language Models

Grigori Perelman — Solving the Unsolvable, Refusing the Prize

Grigori Yakovlevich Perelman (b. 1966)

Origin: Leningrad (now St. Petersburg), Russia

Key work: Proved the Poincaré conjecture (2002–2003, published on arXiv), one of the seven Millennium Prize Problems.

Then **refused** the $1 million prize AND the Fields Medal — the only person to have declined a Fields Medal (as of 2026).

He essentially disappeared from public

Sphere – rubber band contracts

Terence Tao — The Mozart of Math

Terence Chi-Shen Tao (b. 1975)

Origin: Adelaide, Australia (Chinese–Hong Kong descent); works at UCLA

Key work:

- Fields Medal (2006)
- Work spans: harmonic analysis, PDE, analytic number theory, random matrix theory, combinatorics
- Green–Tao theorem (arXiv 2004, published in *Annals of Mathematics* 2008): there are arbitrarily long

Number line with primes highlighted

Maryam Mirzakhani — Breaking Barriers

Maryam Mirzakhani (1977–2017)

Origin: Tehran, Iran; worked at Stanford University

Key work:

- First woman to win the Fields Medal (2014)
- Dynamics and geometry of Riemann surfaces and their moduli spaces
- Connected hyperbolic geometry, complex analysis, and dynamical systems

Maryam Mirzakhani working on large sheets of paper on the floor (Stanford / photographer credit needed)

Genus-0 (sphere)

Karen Uhlenbeck — Geometry and Gauge Theory

Karen Keskulla Uhlenbeck (b. 1942)

Origin: Cleveland, Ohio, USA; worked at University of Texas at Austin and Princeton

Key work:

- First woman to win the Abel Prize (2019)
- "Pioneering achievements in geometric partial differential equations, gauge theory, and integrable systems"
- Work underpins modern mathematical physics, including the

Portrait of Karen Uhlenbeck (Wikimedia Commons)

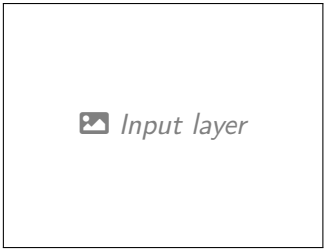
A surface with connections (arrows showing parallel transpor

Machine Learning — The New Mathematics

The mathematics of machine learning:

Math Inside Every AI

- **Linear algebra** — neural networks are matrix operations
- **Calculus** — gradient descent = optimisation via derivatives
- **Probability** — Bayesian inference, statistical learning
- **Information theory** — Shannon's legacy



Key figures:
Geoffrey Hinton (b. 1947, UK/Canada) — deep

The Story Continues

Mathematics is not finished.

Open Problems

- **Riemann hypothesis** (1859) — $1M prize
- **P vs NP** — can every problem whose answer is quickly verifiable also be quickly solved?
- **Navier–Stokes** — do smooth solutions always exist?
- **Birch–Swinnerton-Dyer** conjecture

Main axis

Key Themes — What Have We Learned?

- **Math has no single homeland** — built by people in Central Africa, Babylon, Egypt, Mesoamerica, Greece, India, China, the Islamic world, Europe, and every corner of the globe
- **Mathematical ideas travel** — the same concept was often discovered independently in multiple civilisations
- **People matter** — behind every theorem is a human story
- **Barriers have been broken** — from Hypatia to Noether to Kovalevskaya to Mirzakhani to Uhlenbeck, women have

Simplified world outline

Mathematics was never the achievement of one culture.

It is humanity's collective masterpiece.

Thank You — Questions?

Main axis

What questions do you have?

Which mathematician's story surprised you the most?

Further Reading

Recommended books for curious minds:

- **Journey Through Genius**
  William Dunham
  Great theorems told as human stories
- **The Crest of the Peacock**
  George Gheverghese Joseph
  Non-European roots of mathematics
- **Mathematics in India**
  Kim Plofker
  500 BCE to 1800 CE — the full Indian story
- **A History of Mathematics**
  Victor Katz
  The standard comprehensive textbook

Book covers montage: Journey Through Genius, The Crest of the Peacock, Mathematics in India, A History of Mathematics, The Code Book

QR code linking to a curated online reading